

SE203 – Software Engineering

Introduction

Course Objectives

- This course introduces the different software process models by illustrating its phases and principles of software engineering.
- Topics include:
 - Overview of Software Engineering;
 - Software Process Models;
 - Requirement Engineering Concepts;
 - Software Design;
 - Design Modeling;
 - Software Quality Engineering;
 - Software Project Management;
 - Software Maintenance and Software Evolution.

What is Software?

- Software is a set of items or objects that form a “configuration” that includes
 - programs
 - documents
 - data ...

What is Software?

We can define it as:

- Program (Computer Programs) that when executed provide
 - Desired features
 - Functions and
 - Performance.
- The data on which the program operates is also considered as a part of the software.
- Documents that describe the use of the Programs.

What is Software?

- Software is engineered
- Software doesn't wear out
- Software is complex

Role of Software

A person might be involved with software more than 100 times, often without even realizing that.

We might use a computer

- To check email or weather
- Kitchen appliances
- Traffic signals are now computer controlled
- Mostly we use credit cards in a computerized gasoline pumps.
- Our bank system is highly computerized
- Use of telephones, fax, e-mail or internet, all of these requires Software
- Hotel arrangements will be handled by computers and software
- Airplane & Air traffic control is highly computerized
- Medical Instruments and diagnostic procedures are now computerized

What is the difference between Software and hardware?

Attributes of Good Software

- Should deliver the required functionality and performance to the user.
- It is reliable
- It has good user-interface
- Should be maintainable and reusable

Categories of Computer Software

- System Software
- Application Software
- Engineering/Scientific Software
- Embedded Software
- Web Applications
- Computer Programming tools
- Artificial Intelligence Software

Software--- New Categories

- Ubiquitous computing—wireless networks
- Net sourcing—the Web as a computing engine
- Open source—“free” source code open to the computing community (a blessing, but also a potential curse!)
- Also ... (see Chapter 32)
 - Data mining
 - Grid computing
 - Cognitive machines
 - Software for nanotechnologies

Software Products

Generic products

- Stand-alone systems that are marketed and sold to **any customer** who wishes to buy them.
- Examples – PC software such as editing, graphics programs, project management tools, databases and drawing packages.

Customized products

- Software that is commissioned by **a specific customer** to meet their own needs.
- Examples—systems written to support a particular business process and air traffic control software.

Software Crises

- In the early 60s, software suffered from a problem, which we call the *Software Crisis*
- The techniques that were used to develop small software were not applicable for large software systems.
- In most of the cases the software that was built using the old tools and techniques was not complete.

Software Crises

- Most of the time it was delivered too late.
- Most of the projects were over-budgeted.
- And, in most of the cases, systems built using these techniques were not reliable.
 - **A conference was held in 1960 in which the term “software crisis” was introduced**

Conclusion:

- Software engineering is the result of the software crisis when people realized that it was not possible to construct complex software using the techniques applicable in 1960s.
- An important result of this was that people realized that just coding was not enough; we also need to apply engineering principles.

What is Engineering?

- The process of productive use of scientific knowledge is called Engineering.
- The process of utilizing knowledge and principles to design, build, and analyze objects.

Software Engineering

The science concerned with putting computer science knowledge to practical use.

Software Engineering - IEEE

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.

Computer Science vs. Software Engineering

Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.

System engineering vs. Software Engineering

System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this.

Legacy Software

- *Why must it change?*
 - software must be **adapted** to meet the needs of new computing environments or technology.
 - software must be **enhanced** to implement new business requirements.
 - software must be **extended to make it interoperable** with other more modern systems or databases.
 - software must be **re-architected** to make it viable within a network environment.

Software Costs

- Software costs often dominate computer system costs. The costs of software on a PC are often greater than the hardware cost.
- Software costs more to maintain than it does to develop. For systems with a long life, maintenance costs may be several times development costs.
- Software engineering is concerned with cost-effective software development.

What are the costs of software engineering?

- Roughly **60%** of costs are development costs, **40%** are testing costs. For custom software, evolution costs often exceed development costs.
- Costs vary depending on the type of system being developed and the requirements of system attributes such as performance and system reliability.
- Distribution of costs depends on the development model that is used.

Friendster (2002 – 2011)

- **What was it?**

An early social networking website (before Facebook) where people made profiles, added friends, and joined communities.

- **What went wrong?**

- The site became **very slow** because its **backend database was not scalable**.
- As users increased, the system began to **crash frequently**.
- The development team couldn't fix the engineering issues.

- **Result:**

Users got frustrated and moved to MySpace, and later to Facebook. Friendster shut down in 2011.

2. BlackBerry OS (1999 – 2013)

- **What was it?**

A **mobile operating system** for BlackBerry phones, very popular with business users because of its strong email and security features.

- **What went wrong?**

- When smartphones (iPhone and Android) arrived, people wanted **touchscreens and modern apps**.
- BlackBerry OS was based on older technology and **making apps for it was difficult**.
- The engineering team failed to **modernize or update** it in time.

- **Result:**

Developers stopped building apps, users switched to Android/iOS, and BlackBerry OS died by 2013.

3. Nokia's Symbian OS (1997 – 2014)

- **What was it?**

Nokia's own **mobile operating system**, used in its feature phones and early smartphones. In the 2000s, Nokia was the world's number one phone brand.

- **What went wrong?**

- Symbian's **codebase was very complex**, so updates and maintenance were very slow.
- When iPhone and Android came, they introduced **app stores and modern UI**, which Symbian lacked.
- Developers had **poor tools and support** on Symbian, so new apps stopped coming.

- **Result:**

Symbian looked outdated, users switched to Android/iPhone, and Nokia officially ended Symbian in 2014.

Successful Software Stories

Google Search

Why success?

Highly scalable, extremely fast, constantly optimized algorithms, and strong infrastructure. Continuous innovation (PageRank → AI search).

Facebook (Meta)

Why success?

Engineered to handle billions of users smoothly. Strong scalability + developer tools + continuous feature updates.

Activity

- Analyse the following software and determine how successful or unsuccessful they are. Also, explain the reasons behind their success or failure.
 1. Windows Vista
 2. WhatsApp
 3. Google+
 4. Linux
 5. Microsoft Office
 6. Amazon (E-commerce Platform & AWS)
 7. Yahoo Messenger
 8. Windows Phone OS
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